

Matrices & linear transformations

Defⁿ (Matrix multiplication)

- $A \in M_{m \times n}(F)$

- $v \in F^n$

If $A = [v_1 \ v_2 \ \dots \ v_n]$ } m rows
 $v_1, v_2, \dots, v_n \in F^m$

$$v = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in F^n$$

then define

$$Av = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

1 : ① If A corresponds to linear transformation

$$T: \underline{F^n} \longrightarrow \underline{F^m}$$

we have

$$\underbrace{T(v)}_{m \times 1} = \underbrace{A}_{(m \times n)} \underbrace{v}_{(n \times 1)}$$

(2) We can extend this to a relationship between composition of linear transformations and an operation of multiplication between matrices of the appropriate dimension

(HW 5)

Bases for finite dim'l v. spaces & Matrices

Recall: A basis for
a finite dim'l v. space is
a list $B = (v_1, v_2, \dots, v_n)$
of lin. ind. vectors in V
that span V

Fact: Suppose that V
(HW 5)
is an F -vector space &
have $v_1, v_2, \dots, v_n \in V$
the linear transformation

$$U: F^n \longrightarrow V$$

$$e_i \longmapsto v_i$$

is an isomorphism

$\iff (v_1, v_2, \dots, v_n)$ is
a basis for V

Defⁿ: $T: W \longrightarrow V$ F -linear transformation

is invertible or an isomorphism

if T is bijective

Notation: $T: W \xrightarrow{\sim} V$

Rmk: When T is an isomorphism, the inverse function $T^{-1}: V \rightarrow W$ is also an isomorphism (HW 5)

Slogan: A basis for V is a choice of isomorphism

$$F^n \xrightarrow{\cong} V$$

where $n = \dim(V)$

Observation:

V, W : two finite dim'l
 F -vector spaces

$$T: V \longrightarrow W$$

Suppose that we have
bases $(v_1, \dots, v_n)$ for V

&

$(w_1, \dots, w_m)$ for W

That is, we have isomorphisms

$$\left. \begin{array}{ccc} \underline{U}_1 : F^n & \xrightarrow{\sim} & V \\ \downarrow U_2^{-1} \circ T \circ U_1 & & \downarrow T \\ \underline{U}_2 : F^m & \xrightarrow{\sim} & W \\ & \xleftarrow{U_2^{-1}} & \\ & e_i \longleftarrow w_i & \end{array} \right\}$$

$e_i \longleftarrow w_i$

Now, the linear transformation

$$U_2^{-1} \circ T \circ U_1 : F^n \longrightarrow F^m$$

corresponds to an

$$(m \times n)\text{-matrix } A = [T]_{\beta_1, \beta_2}$$

Therefore, our choices of bases $\beta_1 = (v_1, v_2, \dots, v_n)$ for V

$$\beta_2 = (w_1, w_2, \dots, w_m) \text{ for } W$$

give us a bijection

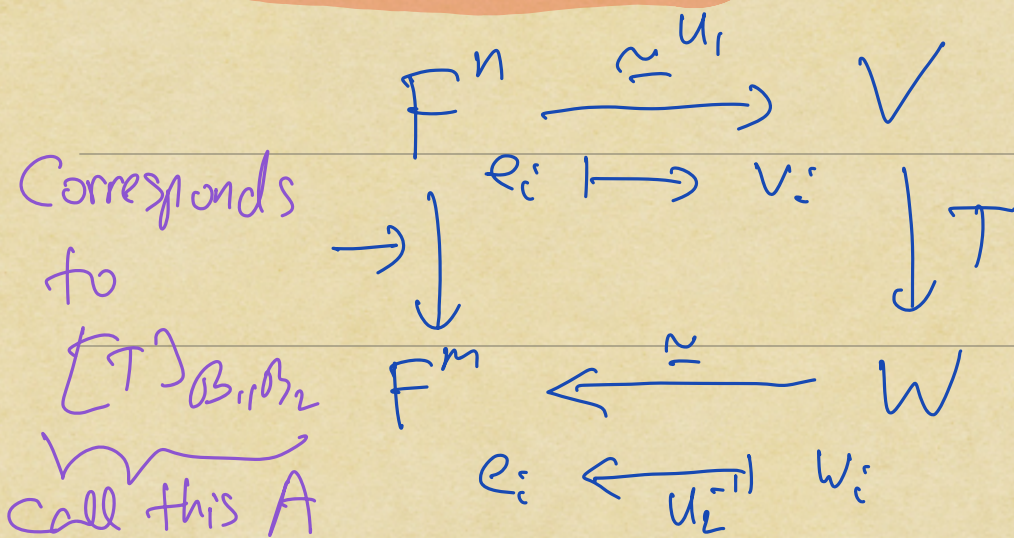
$$\left\{ \begin{array}{l} \text{Linear transformations} \\ T: V \longrightarrow W \end{array} \right\} \longleftrightarrow M_{m \times n}(F)$$

$$T \longmapsto [T]_{\beta_1, \beta_2}$$

Rmk: We can understand linear transformations between finite dim'l vector spaces also in terms of matrices (i.e. not just from F^n to F^m).

But this depends on choices of bases.

Concretely what is $[T]_{\beta_1, \beta_2}$?



Then

$$\begin{aligned} A e_i &= (U_2^{-1} \circ T \circ U_1)(e_i) \\ &= U_2^{-1}(T(U_1(e_i))) \\ &= U_2^{-1}(T(v_i)) \end{aligned}$$

Now

$$T(v_i) \in W$$

|| uniquely

$$a_{1,i} w_1 + a_{2,i} w_2 + \dots + a_{m,i} w_m$$

for $a_{1,i}, a_{2,i}, \dots, a_{m,i} \in F$

$$U_2^{-1}(a_{1,i} w_1 + \dots + a_{m,i} w_m)$$

$$= a_{1,i} U_2^{-1}(w_1) + \dots + a_{m,i} U_2^{-1}(w_m)$$

$$= a_{1,i} e_1 + \dots + a_{m,i} e_m$$

$$A e_i = a_{1,i} e_1 + \dots + a_{m,i} e_m$$

$(\Leftrightarrow)$

$$A e_i = \begin{bmatrix} a_{1,i} \\ a_{2,i} \\ \vdots \\ a_{m,i} \end{bmatrix}$$

where

$$T(v_i) = \sum_{j=1}^m a_{j,i} w_j$$

Example:

$$\begin{array}{ccc} & V & W \\ & \parallel & \parallel \\ D: \text{Poly}_{\leq 2}(\mathbb{R}) & \rightarrow & \text{Poly}_{\leq 1}(\mathbb{R}) \\ f(x) & \mapsto & f'(x) \end{array}$$

$\mathcal{B}_1 = (1, x, x^2)$: basis for V

$\mathcal{B}_2 = (1, x)$: basis for W

$[D]_{\mathcal{B}_1, \mathcal{B}_2}$ is obtained as

follows:

$$\cdot D(1) = 0 = \underline{0} \cdot 1 + \underline{0} \cdot x$$

$$\cdot D(x) = 1 = 1 \cdot 1 + 0 \cdot x$$

$$\cdot D(x^2) = 2x = 0 \cdot 1 + 2 \cdot x$$

$\mathcal{B}_2$
 $\{1, x, x^2\}$

$$[D]_{\beta_1, \beta_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{matrix} \beta_1 \\ \beta_2 \end{matrix}$$